LAB 3: COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Scenario: Alumni Mentorship & Mock Interview Platform

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Architecture Selection

We chose Layered Architecture for the Alumni Mentorship & Mock Interview Platform.

Architectural Choice

The system is divided into separate presentation, business, and data responsibilities. The User Interface handles interaction with students and alumni, the business layer handles mentor matching, session booking, and interview scorecards, and the data layer stores profiles, availability, and feedback information.

Reason 1 – Separation of Concerns

Layered Architecture separates different responsibilities into distinct components. Mentor matching, booking and calendar management, and interview scorecard management can therefore be developed and maintained independently.

Reason 2 – Suitable for System Requirements

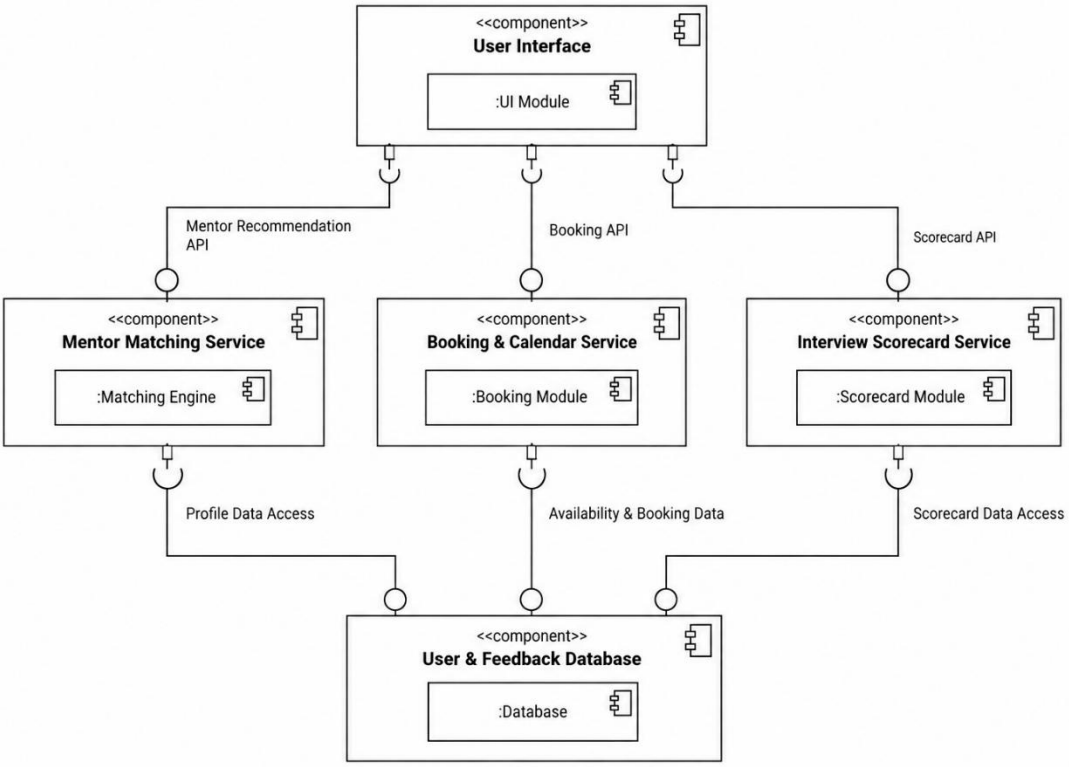
The architecture directly supports the major requirements of the platform, including domain-based mentor recommendations, availability-based session booking, prevention of double booking, and structured post-interview scorecard recording.

Security Advantage

The layered structure prevents the User Interface from directly accessing sensitive profile and feedback data. Access to the database can be controlled through the business layer, helping protect user profiles and interview scorecards as required by the scenario.

Performance Benefit

Separate business components allow operations such as mentor matching and availability checking to be processed efficiently without mixing them with presentation or database logic.


<<component>>
User Interface
:UI Module
Mentor Recommendation API
Booking API
Scorecard API
<<component>>
Mentor Matching Service
:Matching Engine
<<component>>
Booking & Calendar Service
:Booking Module
<<component>>
Interview Scorecard Service
:Scorecard Module
Profile Data Access
Availability & Booking Data
Scorecard Data Access
<<component>>
User & Feedback Database
:Database